Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: _____LO Kai Fung__________

Student ID: _____24021993D__________________

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show ip route ospf

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___Serial0/0/0___

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __129____

Does R2 show up as an OSPF neighbor on R1? ___Yes__

Does R2 show up as an OSPF neighbor on R3? ___No__

What does this information tell you?

R2 S0/0/1 is still passive, so it hasn't formed an adjacency with R3. R3 must route to R2 via R1.

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ___Serial0/0/1___

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65. (Serial link cost 64 + LAN cost 1).

Is R2 listed as an OSPF neighbor to R3? __Yes__

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To let OSPF accurately calculate cost differences for links faster than 100 Mbps 2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Serial link costs became 781. The route to 192.168.3.0/24 is 782 (781 + 1 LAN). The route to 192.168.23.0/30 is 845 (781 + 64 default serial link). 2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562. It is the sum of two 128 kbps serial links (781 + 781). 2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2? 3

The direct link to R3 was manually set to a very high cost (~~1565~~). The longer path through R2 has a lower total cost (1563), so OSPF chose it.

1576

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   To ensure network stability. It prevents the ID from changing and triggering OSPF recalculations if a physical interface goes down.

   4

2. Why is the DR/BDR election process not a concern in this lab?

   Because the routers are connected using point-to-point serial links, which do not require DR/BDR elections.

   4

3. Why would you want to set an OSPF interface to passive?

   To stop sending unnecessary OSPF updates (Hello packets) to LAN networks, which reduces traffic and improves security.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of OSPF Areas (like Area 0) allows large networks to be divided into smaller, manageable zones.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   EIGRP. Routers prefer protocols with the lowest Administrative Distance (AD). EIGRP (AD 90) is chosen over OSPF (110) and RIP (120), ignoring the metrics.

   4